

Monthly Intelligence Report

Edition 003 — Mexico

Where hurricane resilience and seismic preparedness challenge grid stability. edition · June 2026 · SSI v4.0.2

CONTENTS

- [A Executive Summary](#)
- [B Cyber & Economic Exposure Monitor](#)
- [C Data Refresh & Changelog](#)
- [D Deep-Dive: The Gulf Coast–Pacific Corridor](#)
- [E Americas Context Card](#)
- [F SSI Enhanced Neural Network](#)
- [G Looking Ahead](#)

SECTION A

Executive Summary

SUBSTATIONS SCORED

3,140

HV + MV across 32 Estados

GRID LINES MAPPED

~30,396

~30,000 km transmission & distribution

MC SIMULATIONS

~31M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~31 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 3,140 substations and ~30,396 grid lines spanning ~30,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (CENACE, CRE, CFE, INEGI, SMN, CENAPRED), making the methodology fully auditable and reproducible. Initially covering Mexico's CENACE-managed transmission and ~250 CFE distribution divisions (DSOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Sub 677812521

Baja California, Baja California · 230 kV

0.324

Medium

0.086

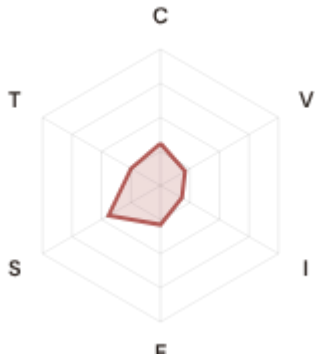
8th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.309 / 0.30 (103%)
I Infrastructure: 0.182 / 0.25 (73%)
S Saturation: 0.440 / 0.20 (220%)
V Voltage: 0.208 / 0.10 (209%)
E Economic: 0.288 / 0.10 (288%)
T Transition: 0.250 / 0.05 (500%)

MODIFIERS

R3 Consequence: 1.013 ↑ amplifies
R4 Graph Criticality: 1.065 ↑ amplifies
R6a Restoration: 1.017 ↑ amplifies
R6b Seismic: 1.010 → neutral
R7 Digital Readiness: 1.013 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Sub 677812521** in Baja California, Baja California. With an R_final of 0.324 (Medium band, 8th fleet percentile), its risk profile is dominated by the T (Transition) component at 500% of its weight ceiling, followed by E (Economic) at 288%. Modifiers collectively shift R_base by 12.3%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Mexico's Energy Regulatory Commission (CRE) and CENACE network operator strategic planning requires.

CYBER HIGH EXPOSURE

0
0.0% of fleet

BLIND SPOTS

88
High/Critical + R7 < median

AVG R7 MODIFIER

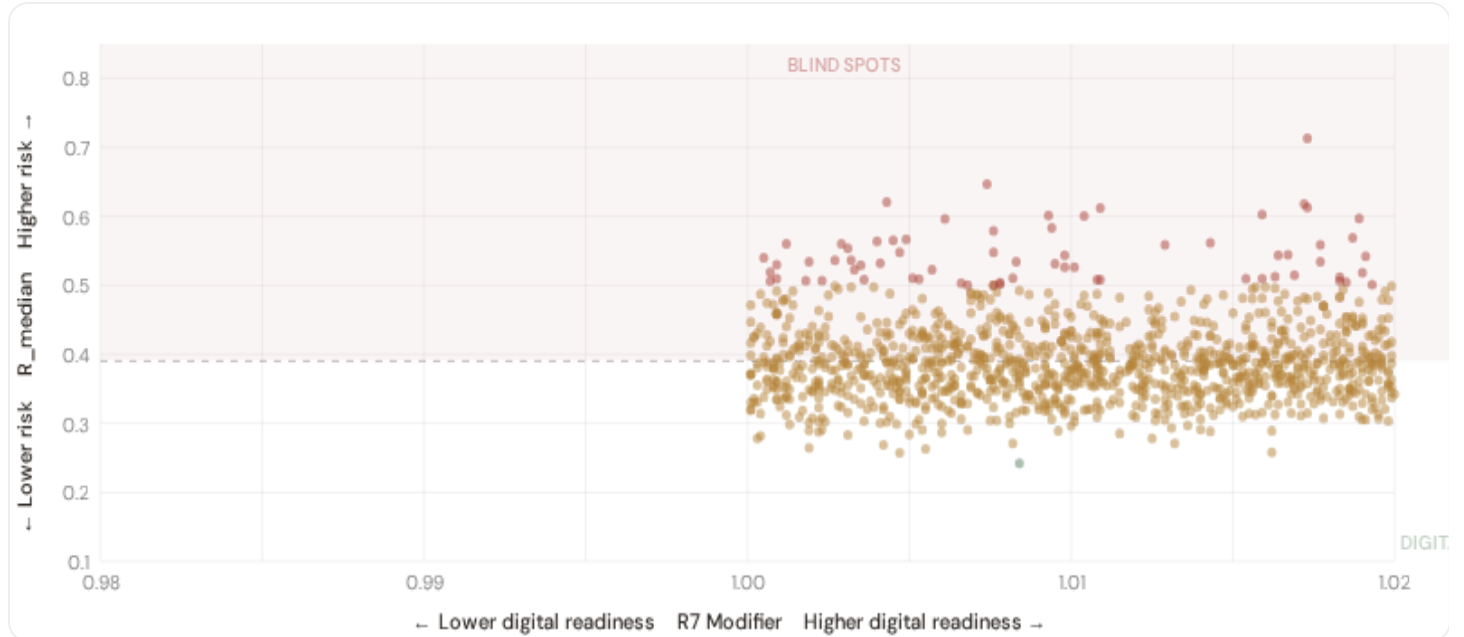
1.0257
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

775

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



The cyber-physical matrix identifies **88 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0260$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Chiapas (26), Oaxaca (21), Guerrero (15)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Capital-Intensive / High-Consequence

Est. VoLL: Mex\$300–600/kWh (MXN)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

775 substations (24.7%) High/Critical: **81** (10.5%) Avg R: **0.419** Avg E: **0.325** / 0.10

Low 0

Med 694

High 81

Crit 0

Industrial / Medium–Large Enterprise

Est. VoLL: Mex\$150–300/kWh (MXN)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

802 substations (25.5%) High/Critical: **50** (6.2%) Avg R: **0.404** Avg E: **0.323** / 0.10



Commercial / SME–Dense

Est. VoLL: Mex\$60–150/kWh (MXN)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

775 substations (24.7%) High/Critical: **39** (5.0%) Avg R: **0.396** Avg E: **0.324** / 0.10



Light / Agricultural / Rural

Est. VoLL: Mex\$20–60/kWh (MXN)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

788 substations (25.1%) High/Critical: **28** (3.6%) Avg R: **0.372** Avg E: **0.323** / 0.10

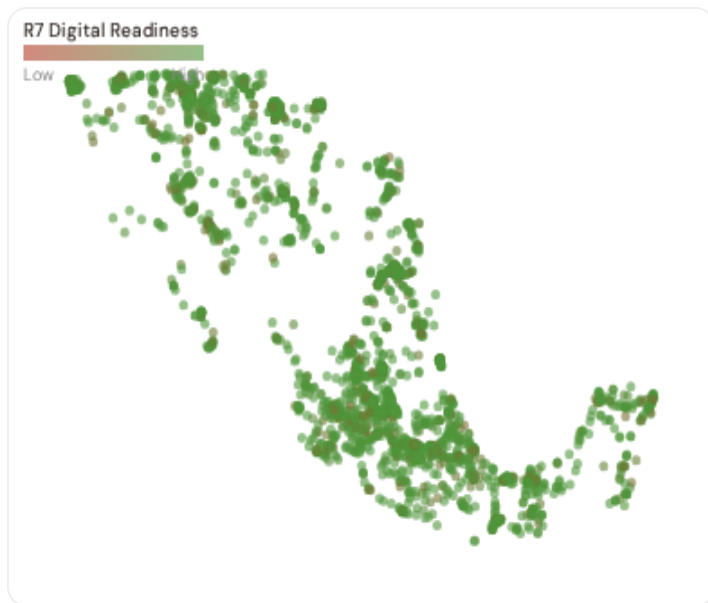


Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **81 substations** combine capital-intensive economic fabric ($R3 \geq 1.07$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Sonora (121), Chihuahua (90), Nuevo León (78)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 32.0%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Estados & Municipios Digital Readiness Ranking

Estados ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_{median} to identify regions combining digital exposure with high grid stress. Reference CENACE/CRE digital infrastructure investments and broadband penetration gaps in southern states.



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Durango		1.0087
2	Ciudad de México		1.0222
3	Yucatán ▲ high R		1.0224
4	Quintana Roo ▲ high R		1.0230
5	Baja California Sur		1.0231
6	Chiapas ▲ high R		1.0235
7	Sinaloa		1.0235
8	Veracruz ▲ high R		1.0242
9	Tlaxcala		1.0242
10	Guanajuato		1.0246
11	Tabasco ▲ high R		1.0252
12	Jalisco ▲ high R		1.0253
13	Baja California		1.0254
14	Puebla ▲ high R		1.0255
15	San Luis Potosí		1.0255

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Durango** has the lowest average R7 (1.0087), while **Campeche** leads at 1.0292 — a spread of 0.0206 across the R7 range. **9 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0257, median = 1.0260; 0 substations classified HIGH cyber-exposure; 88 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **25 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

25

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

16

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

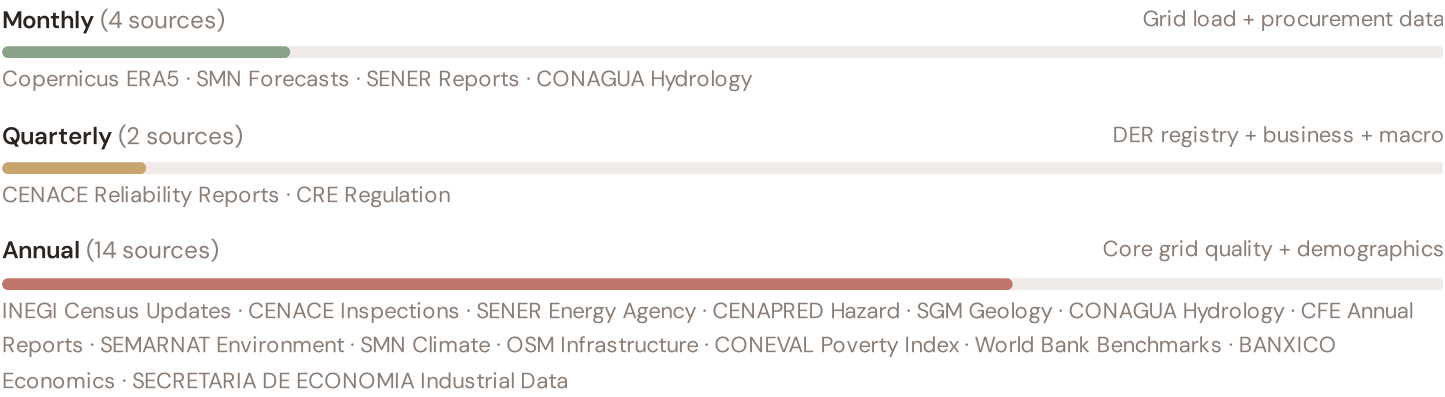
Data Source Registry — June 2026

SMN	SMN — Servicio Meteorológico Nacional	DAILY	Station	8 vars
CFE	CFE — Comisión Federal de Electricidad	MONTHLY	Distribution Region	9 vars
CONAGUA	CONAGUA — Comisión Nacional del Agua	MONTHLY	Watershed	5 vars
OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
PEMEX	PEMEX — Petróleos Mexicanos	MONTHLY	Regional	3 vars
CRE	CRE — Comisión Reguladora de Energía	QUARTERLY	DSO	12 vars
Banxico	Banco de México — Central Bank	QUARTERLY	National	3 vars
INEGI	INEGI — Instituto Nacional de Estadística y Geografía	ANNUAL	Municipio	10 vars
CENAPRED	CENAPRED — Centro Nacional de Prevención de Desastres	ANNUAL	Grid 0.1°	7 vars
SENER	SENER — Secretaría de Energía	ANNUAL	National	4 vars
SEMARNAT	SEMARNAT — Secretaría de Medio Ambiente	ANNUAL	Municipio	5 vars
CONUEE	CONUEE — Comisión Nacional para el Uso Eficiente de la Energía	ANNUAL	Regional	3 vars
SGM	SGM — Servicio Geológico Mexicano	ANNUAL	Grid 0.1°	6 vars
PRODESEN	PRODESEN — Programa de Desarrollo del SEN (Grid Planning)	ANNUAL	Regional	4 vars
INEEL	INEEL — Instituto Nacional de Electricidad y Energías Limpias	ANNUAL	Regional	3 vars
IMTA	IMTA — Instituto Mexicano de Tecnología del Agua	ANNUAL	Watershed	3 vars
SSA	SSA — Secretaría de Salud	ANNUAL	Municipio	2 vars
CONAPO	CONAPO — Consejo Nacional de Población	ANNUAL	Municipio	3 vars
SAT	SAT — Servicio de Administración Tributaria	ANNUAL	Municipio	2 vars

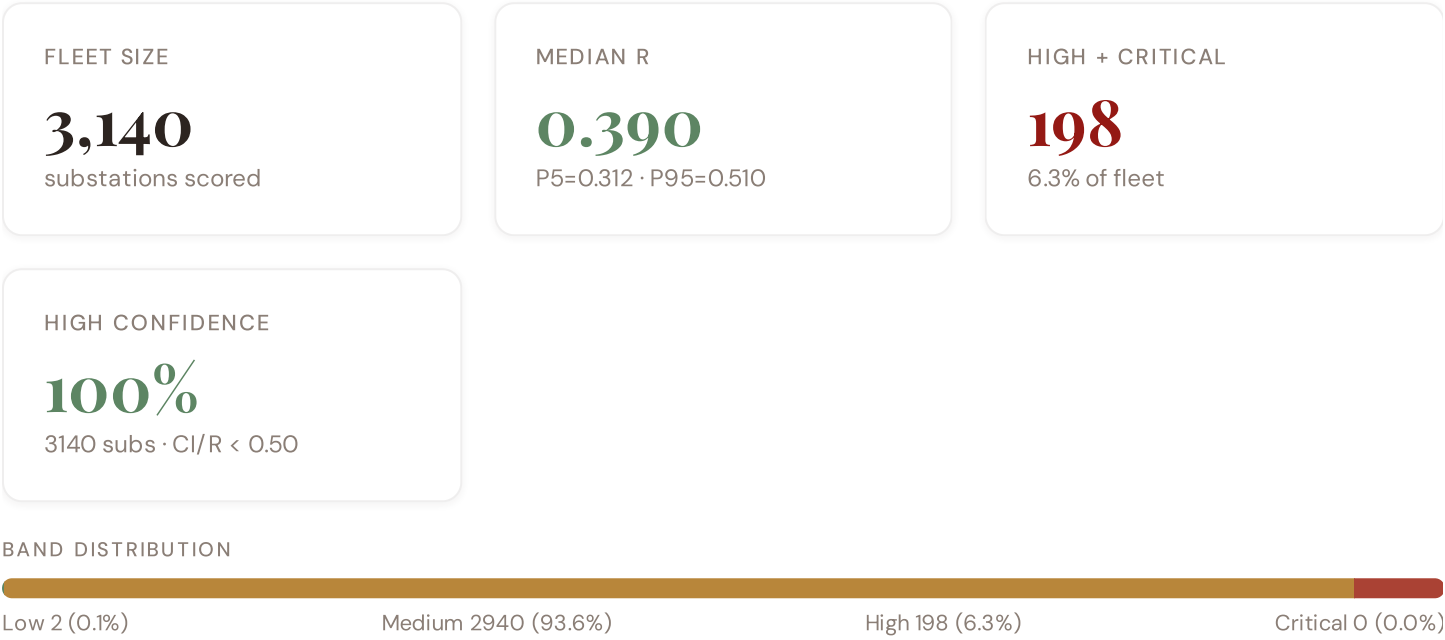
ASEA	ASEA — Agencia de Seguridad, Energía y Ambiente	ANNUAL	Infrastructure Site	2 vars
SE	SE — Secretaría de Economía	ANNUAL	Municipio	2 vars
CENACE	CENACE — Centro Nacional de Control de Energía	REAL-TIME	Substation	14 vars
CONEVAL	CONEVAL — Consejo Nacional de Evaluación	BIENNIAL	Municipio	4 vars
AEMO_CFE	AEMO/CFE Generation Data — Capacity + Output	REAL-TIME	Power Plant	5 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **3,140 substation scores remain unchanged** this edition. The fleet median R of 0.390 (mean 0.398) reflects a distribution skewed toward the lower bands, with 0.1% in Low and 93.6% in Medium. The first material score changes are expected when the national regulator publishes

the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

15 changes tracked across PO, P1, and P2 releases

MX1	NEW	Mexico country launch — 3,140 substations across 32 Estados, administrative municipio level
MX2	DATA	Integrated CENACE transmission dispatch data + CRE reliability benchmarks
MX3	DATA	CRE reliability register integrated — SAIDI/SAIFI standardisation for CFE distribution and permit holders
MX4	DATA	SMN tropical cyclone + rainfall hazard data integrated — critical for maritime climate exposure assessment
MX5	DATA	CENAPRED seismic + volcanic + flood hazard mapping integrated — earthquake, volcanic, flood zone vulnerability
MX6	DATA	SGM wind loading + geological stability mapping integrated — overhead transmission line vulnerability in seismic regions
MX7	DATA	SMN + CENAPRED + SGM climate integration — tropical cyclone risk, seismic hazard, flood exposure, volcanic activity
MX8	DATA	INEGI Census + CONEVAL socio-economic data integrated — vulnerability indices per municipio, urban concentration
MX9	DATA	SENER + CENACE data integrated — RE penetration for northern corridor, solar expansion, PRODESEN targets
MX10	NEW	Banxico + regional market data integrated — E1 volatility modulation for RE-dependent regions
MX11	ENHANCED	R6b modifier enhanced with triple SMN+CENAPRED+CONAGUA hazard overlay: tropical cyclone + seismic/volcanic + flood (range 1.00–1.25)
MX12	ENHANCED	R3 consequence enriched with urban concentration, socio-economic vulnerability, poverty indices + industrial exposure metrics
MX13	ENHANCED	R4 graph criticality rebuilt with CENACE transmission topology + CFE network constraints + regional dispatch centre bottleneck penalty
MX14	NEW	I9 Renewable Capacity Concentration Risk — new metric from SENER/CENACE for major wind/solar farm proximity (northern corridor growth)
MX15	ENHANCED	R2 Adaptive IRI now includes climate projections for tropical cyclone intensity, seismic activity, flood risk shifts

Deep-Dive: The Gulf Coast–Pacific Corridor

The Gulf Coast–Pacific Corridor — spanning Veracruz, Guerrero, Oaxaca, Tabasco, and surrounding estados — faces a uniquely compounded hazard environment that makes grid resilience particularly challenging. This geographic corridor experiences both Atlantic hurricane season (August–October) and simultaneous Pacific hurricane season (May–November), creating a six-month window of dual storm exposure. Overlaid on this are three major seismic fault systems (Mexico–Cocos boundary, Venta Rica fault, and Tehuantepec Ridge), plus ongoing geothermal stress in volcanic regions (Trans–Mexican Belt). The convergence of these natural hazards — each independently capable of causing cascading failures — makes the Corridor a critical case study in the SSI framework.

The R6b modifier is weighted especially heavily in this geography, using SMN hurricane historical reanalysis and CENAPRED seismic scenario models to capture the interaction of compound hazards with substation topology and restoration capacity. CENACE's post-2019 hardening investments have focused on this corridor, recognizing that a simultaneous multi-state outage (triggered by a major hurricane–seismic coupled event) would ripple across the entire Baja California–Texas cross-border power exchange and disrupt North American manufacturing supply chains dependent on Mexican energy stability.

Corridor-level analysis reveals substations where high exposure to compound natural hazards intersects with limited restoration pathways. The SSI's asset-level resolution identifies critical nodes where grid hardening investment yields maximum resilience return — a strategic advantage over traditional regional blackout risk models which operate at lower resolution.

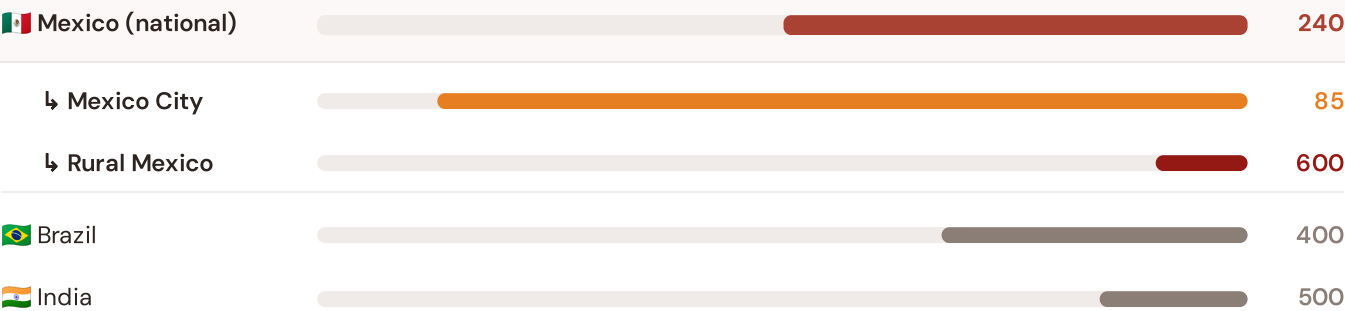
SECTION E — RECURRING MODULE

Americas Context Card

How this works: Each edition includes one Americas Context Card drawn from the \$16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Gulf Coast–Pacific corridor analysis hinges on Mexico's continuity performance relative to international peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Colombia (003), CMIP6 climate hazard vs Caribbean peers (004), etc. CRE reliability reports, CENACE operational data, and CFE distribution metrics update annually with regulatory filings.

Unplanned SAIDI (min/customer/year) — Mexico vs International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year (excl. major events).



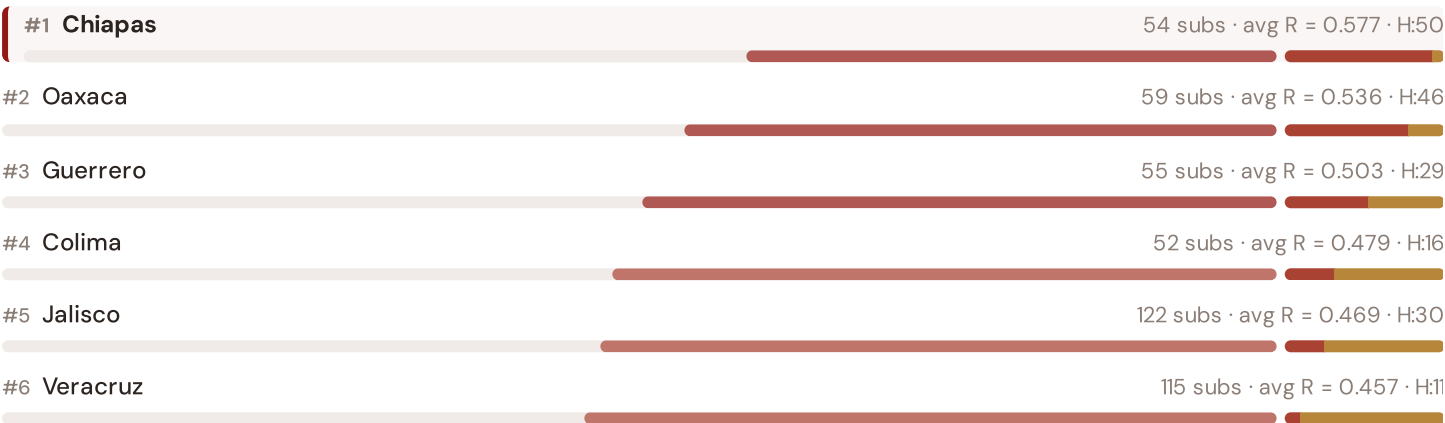
Source: CRE reliability reports (2023–2024) for Mexico; CENACE transmission grid operations; CFE distribution metrics; EIA Form 861 (2023) for US/Canada; IEA statistics for international peers. Mexico values exclude major event days (CRE methodology). · See §16 Peer Benchmarking for full methodology. · Updated annually.

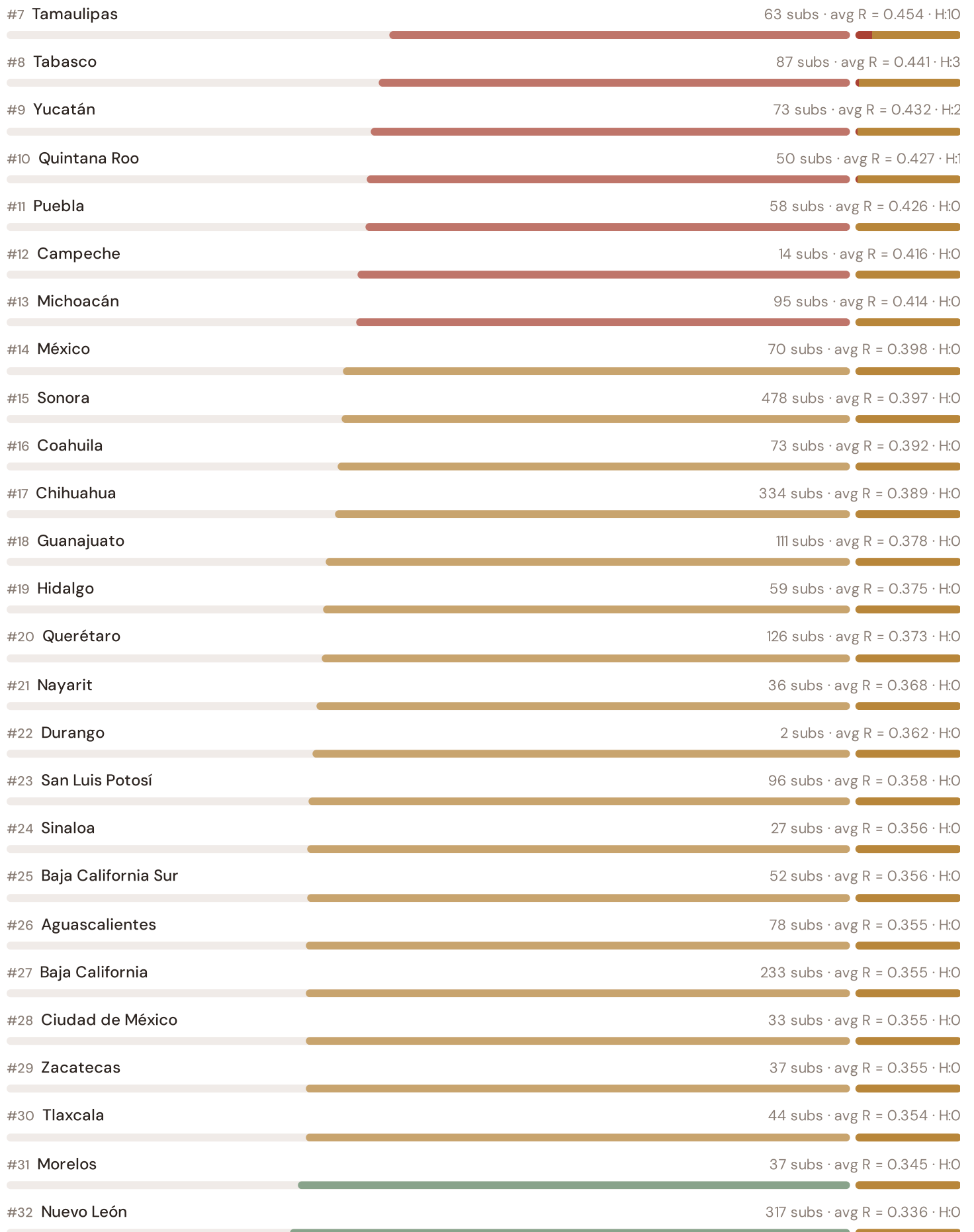
This month's key insight: The Chiapas corridor deep-dive shows **54 substations** (of 54) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.321 is **1.0× the fleet average** of 0.331. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 32 Estados

Where does the Gulf Coast–Pacific corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





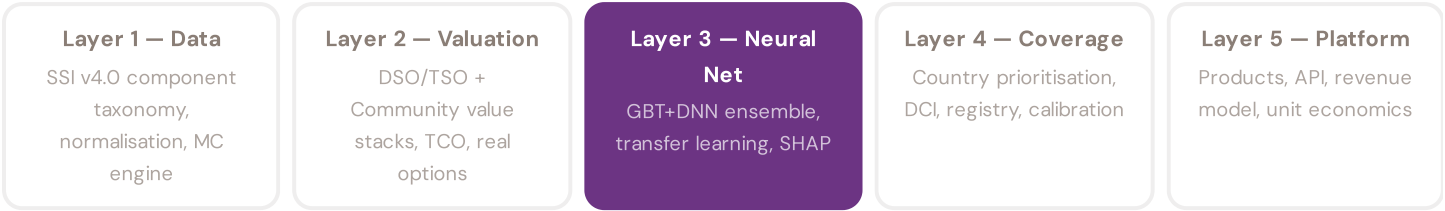
The Chiapas ranks **#1 of 32 region regions** by mean R_{median}, placing it among the upper tier of national risk. The three most stressed are Chiapas, Oaxaca, Guerrero, while the three most resilient are Nuevo León, Morelos, Tlaxcala. 14 of 32 score above the fleet average of 0.398. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is

precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to battery storage valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery storage system worth at this substation — to CENACE/CRE and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics for Mexico's energy infrastructure.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 3,140 substations with complete feature vectors. Average CI_width of 0.104 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 2 Low, 2940 Medium, 198 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

CENACE Reliability Reports, **CRE Regulation** are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **INEGI — Instituto Nacional de Estadística y Geografía** (10 variables → C1–C4, E2–E3, demographics, population served per municipio, income distribution, regional economic data, census data).

FLEET SPOTLIGHT

Chiapas: Highest Risk Concentration

Chiapas has the highest concentration of High/Critical substations: **50 of 54 (93%)** are in the upper risk bands — avg R = 0.577. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Mexican utility, CENACE, CRE, CFE distribution division, estado government, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI v4.0.2 · 3,140 substations · 30,396 power lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5

© 2026 Altinium Invest · Corso Vittorio Emanuele 110/5, 80121 Napoli · P.IVA 10656441218

USCO Registration pending · UK IPO Registration · Protected under CDPA 1988, Berne Convention, Copyright Act 1976

[View in browser](#)